

Análisis de la etapa PosePrediction

1. Objetivos

OBJ1. Análisis de la etapa Sw PosePrediction para identificación de opciones de mejora de rendimiento

OBJ2.

2. Visión general y estrategias

La medida de tiempos de ejecución del pipeline completo arroja los resultados de la tabla, donde se puede ver que la etapa Tracking:: PosePrediction se ha convertido, tras aplicar el coprocesador, en el cuello de botella con 71.67ms

pipe	etapa	Zynq UltraScale+ (Petalinux) 4x cortexA53@1,2 GHz, L1 32K, L2 1MB(shared)			
		1Ghz Sw only	1Ghz copro	1.3Ghz Sw only	1.3Ghz copro
CAPTURA	Captura	-	-	-	-
	Acond.	-	-	-	-
	Color conv.	-	-	-	-
	Escalado	-	-	-	-
	VDMA	-	-	-	-
	Total	13.8-17.3	13.8-17.3	13.8-17.3	13.8-17.3
TRACKING	ORB	171.95	27.75	135.07	27.75
	Octree	-	-	-	-
	Pose Pred.	76.38	102.15	63.58	71.67
	LM track	13.46	19.76	10.08	13.57
	KF decision	0.35	0.39	0.33	0.29
	Total	262.14	147.55	209.06	107.27

Para intentar reducir este tiempo se va a realizar un profiling de la etapa PosePrediction identificando las funciones más costosas computacionalmente. A continuación se estudiará la viabilidad de acelerarlas mediante 3 estrategias:

- Utilización del coprocesador NEON de los ARM
- Aproximación multihebra con OMP
- Aceleración por Hw

3. Profiling

• Análisis dinámico

La idea es utilizar una herramienta de profiling tipo Gprof para hacer profiling dinámico https://ftp.gnu.org/old-gnu/Manuals/gprof-2.9.1/html_mono/gprof.html

para encontrar las funciones que suponen el cuello de botella de la etapa (flat profile, call graph). Esto supone recompilar el código y linkarlo con las libs de gprof.

Para ello se clonó el repositorio DAA y se intentó recompilar sin utilizar las VLIBS. Aunque se consiguió recompilar, las imágenes no se llegan a procesar correctamente y no es posible obtener la información necesaria.

Para conseguir la aplicación completamente funcional es necesario incluir las VLIBS, lo que supone tener los permisos necesarios.

Se decidió, postergar este análisis hasta que Víctor tenga preparado el docker para compilación cruzada y realizar el profiling sobre la XCU (en este caso se utilizan las VLIBS ya compiladas que se enlazan de forma estática).

- **Análisis estático del código**

La hipótesis es que el cuello de botella son las funciones del flujo de ejecución **mpSLAM->TrackMonocular** donde se llama a:

```
g2o::OptimizationAlgorithmLevenberg(solver_ptr)
```

Son dos funciones:

```
// file: Optimizer.cc
```

```
int Optimizer::PoseOptimizationGPS(Frame *pFrame){
    solver = new g2o::OptimizationAlgorithmLevenberg(solver_ptr);
    optimizer.setAlgorithm(solver);

    for(size_t it=0; it<4; ++it) {
        optimizer.optimize(its[it]);
    }
}

int Optimizer::PoseOptimization(Frame *pFrame){
    solver = new g2o::OptimizationAlgorithmLevenberg(solver_ptr);
    optimizer.setAlgorithm(solver);

    for(size_t it=0; it<4; ++it) {
        optimizer.optimize(its[it]);
    }
}
```

Nota: También se llama dese otras funciones (BundleAdjustmentGPS, BundleAdjustment, FullInertialBA, LocalBundleAdjustment, LocalBundleAdjustmentGPS, OptimizeEssentialGraph, OptimizeSim3, LocalInertialBA, InertialOptimization, MergeInertialBA, OptimizeEssentialGraph4DoF) pero no involucradas en el mpSLAM->TrackMonocular

Estas funciones son a su vez llamadas desde las siguientes funciones de Tracking.cc:

a) PoseOptimizationGPS

- TrackWithGPSData
- TrackLocalMap (no está en el código de PosePrediction)

b) PoseOptimization

- TrackReferenceKeyFrame
- TrackWithGPSData
- TrackWithMotionModel
- TrackLocalMap
- Relocalization
- Relocalization2

El código correspondiente a **Pose_Prediction** define 2 flujos de ejecución:

MAPEO y **LOCATION**

A continuación se muestra el código con las funciones críticas en negrita

// FILE: Tracking.cc

```
#ifdef REGISTER_TIMES
    std::chrono::steady_clock::time_point time_StartPosePred ...
#endif
// Initial cam pose estimation using motion model or relocalization (if tracking is lost)
2062 if(!mbOnlyTracking) { // OBJETIVO: MAPEO
    if(mState==OK) {
        //LAST GPS CALCULATIONS
        // Local Mapping might have changed some MapPoints tracked in last frame
        CheckReplacedInLastFrame();
        if(mapgps) {
            bOK = TrackWithGPSData(); // -- > PoseOptimizationGPS
            if(!bOK)
                bOK = TrackReferenceKeyFrameGPS();
        }
        else if(!mbVelocity && ....) {
            bOK = TrackReferenceKeyFrame(); // -- > PoseOptimization
        }
        else {
            bOK = :();
            if(!bOK)
                bOK = TrackReferenceKeyFrame(); // -- > PoseOptimization
        }
        if (!bOK){ }
    }
    else {
        if (mState == RECENTLY_LOST) {
            if((mSensor == System::IMU_MONOCULAR || ....) {
                if(pCurrentMap->isImuInitialized()){
                    PredictStateIMU();
                }
                else{ }
                if (mCurrentFrame.mTimeStamp-....) { }
            }
            else { // Relocalization
                if(!mapgps || mbOnlyTracking)
                    Relocalization2(); // -- > PoseOptimization
            }
            if(mCurrentFrame.mTimeStamp-mTimeS.) { }
        }
        else if (mState == LOST) { }
    }
}
```

```

2193     else // OBJETIVO: LOCATION
        {
            // Localization Mode: Local Mapping is deactivated
            if(mState==LOST) {
                bOK = Relocalization2(); // -- > PoseOptimization
            }
            else {
                if(!mbVO) { //mbGPS
                    if(mapgps) {
                        bOK = TrackWithGPSData();
                    }
                    else if(mbVelocity) { // In last frame we tracked enough MapPoints
                        bOK = TrackWithMotionModel(); // -- > PoseOptimization
                    }
                    else {
                        bOK = TrackReferenceKeyFrame(); // -- > PoseOptimization
                    }
                }
                else { // In last frame we tracked mainly "visual odometry" points.
                    if(mapgps){ // IS NOT EXECUTED SLAM MODE
                        TrackWithGPSData(); // -- > PoseOptimization
                        TcwMM = mCurrentFrame.GetPose();
                    }
                    else if(mbVelocity) {
                        bOKMM = TrackWithMotionModel(); // -- > PoseOptimization
                        TcwMM = mCurrentFrame.GetPose();
                    }
                    bOKReloc = Relocalization2();
                    if(bOKMM && !bOKReloc) {
                        mCurrentFrame.SetPose(TcwMM);
                    }
                }
            }
        }
2285 }

```

```

#ifdef REGISTER_TIMES
    std::chrono::steady_clock::time_point time_EndPosePred ...
#endif

```

A) Confirmación del cuello de botella.

Se ha instrumentado el flujo de ejecución MAPEO

- Tracking.cc / Tracking.h
- CMakeList.txt

obteniéndose:

Versión original (-O3 full, non cross): Tracking

ORB Extraction: 148.32004\$ \pm \$10.80446

Pose Prediction: 77.69082\$ \pm \$77.09461

LM Track: 12.89606\$\pm\$12.14011
New KF decision: 0.33789\$\pm\$0.40301
Total Tracking: nan\$\pm\$nan

Versión MAPEO (-O3 full, non cross):Tracking

ORB Extraction: 148.38\$\pm\$10.715
Pose Prediction: 74.624\$\pm\$73.007
MAPEO->ChkRpllnLstFrm: 0.20612\$\pm\$0.12619
MAPEO->TrkWithGPSDat : 99.197\$\pm\$67.908
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 12.681\$\pm\$11.843
New KF decision: 0.31109\$\pm\$0.3988
Total Tracking: nan\$\pm\$nan

Tracking

ORB Extraction: 147.61\$\pm\$10.832
Pose Prediction: 77.809\$\pm\$64.057
MAPEO->ChkRpllnLstFrm: 0.18131\$\pm\$0.11612
MAPEO->TrkWithGPSDat : 88.069\$\pm\$60.972
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 12.751\$\pm\$10.243
New KF decision: 0.33055\$\pm\$0.37461
Total Tracking: nan\$\pm\$nan

Versión MAPEO (-O3 full, non cross):Tracking

MAPEO→TrkWithGPSDat→ PoseOptimization

MAPEO→TrkWithGPSDat→ PoseOptimizationGPS

Versión LOCALIZACIÓN (-O3 full, non cross):Tracking

Observaciones:

- TrackWithGPSData() parece ser el cuello de botella. Esta función llama a ambas funciones críticas: PoseOptimizationGPS /PoseOptimization.
- Cuando la ejecución pasa por TrackWithGPSData() los tiempos de MAPEO son muy altos y sube la media de PosePrediction

TODO: evaluar el flujo LOCALIZACIÓN

```
bool Tracking::TrackWithGPSData()
{
    // Update last frame pose according to its reference keyframe
    // Create "visual odometry" points if in Localization Mode
    //if(mbOnlyTracking) UpdateLastFrame();

    // Header
    if (mpAtlas->isImuInitialized() && ... ) {
    }
    else{
    }
    fill(mCurrentFrame.mvpMapPoints.begin(),mCurrentFrame.mvpMapPoints.end(),...);
    int nmatches = matcher.SearchByProjection(mCurrentFrame,mLastFrame,...);
    // If few matches, uses a wider window search
    if(nmatches<20) {
        fill(mCurrentFrame.mvpMapPoints.begin(),mCurrentFrame.mvpMapPoints.end(),...);
        nmatches = matcher.SearchByProjection(mCurrentFrame,mLastFrame,...);
    }
    if(nmatches<20 && !mbOnlyTracking){//20
    }
    // Optimize frame pose with all matches
    if(!mbOnlyTracking)
        Optimizer::PoseOptimizationGPS(&mCurrentFrame); --> optimizer.optimize(its[it]);
    else
        Optimizer::PoseOptimization(&mCurrentFrame);
    // Discard Outliers
    int nmatchesMap = 0;
    for(int i =0; i<mCurrentFrame.N; i++)
    {
    }
    if(mbOnlyTracking)
    {
    }
}
```

ejecución_1

ORB Extraction: 147.73\$\pm\$11.289

Pose Prediction: 77.396\$\pm\$69.928

MAPEO->ChkRplInLstFrm: 0.197\$\pm\$0.11853

MAPEO->TrkWithGPSDat : 92.481\$\pm\$66.495

TrkWithGPSDat->Header : 95.677\$\pm\$55.483

TrkWithGPSDat->Opt : 10.711\$\pm\$6.0632

TrkWithGPSDat->Outl : 0.10964\$\pm\$0.062792

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 12.934\$\pm\$10.722

New KF decision: 0.3187\$\pm\$0.36882

Total Tracking: nan\$\pm\$nan

ejecución_2

ORB Extraction: 147.59\$\pm\$11.305

Pose Prediction: 123.45\$\pm\$99.741

MAPEO->ChkRplInLstFrm: 0.28779\$\pm\$0.15372

MAPEO->TrkWithGPSDat : 146.56\$\pm\$91.412

TrkWithGPSDat->Header : 144.68\$\pm\$76.243

TrkWithGPSDat->Opt : 15.805\$\pm\$7.9995

TrkWithGPSDat->Outl : 0.17366\$\pm\$0.092677

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 21.281\$\pm\$15.82

New KF decision: 0.42884\$\pm\$0.42122

Total Tracking: nan\$\pm\$nan

ejecución_3

ORB Extraction: 148.53\$\pm\$10.899

Pose Prediction: 104.79\$\pm\$75.112

MAPEO->ChkRplInLstFrm: 0.25491\$\pm\$0.11395

MAPEO->TrkWithGPSDat : 124.62\$\pm\$64.773

TrkWithGPSDat->Header : 114.2\$\pm\$57.309

TrkWithGPSDat->Opt : 12.51\$\pm\$6.4038

TrkWithGPSDat->Outl : 0.13021\$\pm\$0.067154

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 17.55\$\pm\$11.794

New KF decision: 0.38671\$\pm\$0.39516

Total Tracking: nan\$\pm\$nan

```

bool Tracking::TrackWithGPSData()
{
    // Update last frame pose according to its reference keyframe
    // Create "visual odometry" points if in Localization Mode
    //if(mbOnlyTracking) UpdateLastFrame();

    // Header
    if (mpAtlas->isImuInitialized() && ... ) {
    }
    else{
    }
    fill(mCurrentFrame.mvpMapPoints.begin(),mCurrentFrame.mvpMapPoints.end(),...);
    // HeaderSrch
    int nmatches = matcher.SearchByProjection(mCurrentFrame,mLastFrame,...);
    // If few matches, uses a wider window search
    if(nmatches<20) {
        fill(mCurrentFrame.mvpMapPoints.begin(),mCurrentFrame.mvpMapPoints.end(),...);
        nmatches = matcher.SearchByProjection(mCurrentFrame,mLastFrame,...);
    }
    if(nmatches<20 && !mbOnlyTracking){//20
    }
    // Optimize frame pose with all matches
    if(!mbOnlyTracking)
        Optimizer::PoseOptimizationGPS(&mCurrentFrame); --> optimizer.optimize(its[it]);
    else
        Optimizer::PoseOptimization(&mCurrentFrame);
    // Discard Outliers
    int nmatchesMap = 0;
    for(int i =0; i<mCurrentFrame.N; i++)
    {
    }
    if(mbOnlyTracking)
    {
    }
}

```

Original

ORB Extraction: 148.15\$\pm\$10.752

Pose Prediction: 75.486\$\pm\$62.979

MAPEO->ChkRplInLstFrm: 0.17772\$\pm\$0.10824

MAPEO->TrkWithGPSDat : 87.048\$\pm\$59.521

TrkWithGPSDat->Header : 82.745\$\pm\$51.98

TrkWithGPSDat->HeaderSrch : 78.442\$\pm\$53.692

TrkWithGPSDat->Opt : 8.7621\$\pm\$6.0904

TrkWithGPSDat->Outl : 0.091389\$\pm\$0.058538

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 12.238\$\pm\$9.9341

New KF decision: 0.30609\$\pm\$0.36392

Total Tracking: nan\$\pm\$nan

Varias ejecuciones seguidas:

Ejecución 1:

ORB Extraction: 147.21\$\pm\$11.264
Pose Prediction: **41.765**\$\pm\$42.369
MAPEO->ChkRplInLstFrm: 0.17765\$\pm\$0.090256
MAPEO->TrkWithGPSDat : 62.635\$\pm\$37.013
 TrkWithGPSDat->Header : 58.095\$\pm\$31.329
 TrkWithGPSDat->HeaderSrch : **54.97**\$\pm\$33.058
 TrkWithGPSDat->Opt : 7.8422\$\pm\$4.0282
 TrkWithGPSDat->Outl : 0.081912\$\pm\$0.046368
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.3893\$\pm\$8.0892
New KF decision: 0.2541\$\pm\$0.33483
Total Tracking: nan\$\pm\$nan

Ejecución 2:

ORB Extraction: 146.59\$\pm\$11.656
Pose Prediction: **13.414**\$\pm\$21.029
MAPEO->ChkRplInLstFrm: 0.09282\$\pm\$0.060177
MAPEO->TrkWithGPSDat : 27.46\$\pm\$22.705
 TrkWithGPSDat->Header : **30.122**\$\pm\$19.195
 TrkWithGPSDat->HeaderSrch : 22.492\$\pm\$20.433
 TrkWithGPSDat->Opt : 4.8956\$\pm\$3.1517
 TrkWithGPSDat->Outl : 0.050959\$\pm\$0.023026
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 2.7675\$\pm\$4.3062
New KF decision: 0.14387\$\pm\$0.29545
Total Tracking: nan\$\pm\$nan

Ejecución 3:

ORB Extraction: 144.13\$\pm\$11.64
Pose Prediction: **41.354**\$\pm\$43.985
MAPEO->ChkRplInLstFrm: 0.17029\$\pm\$0.0893
MAPEO->TrkWithGPSDat : 62.129\$\pm\$39.999
 TrkWithGPSDat->Header : 57.698\$\pm\$34.125
 TrkWithGPSDat->HeaderSrch : **54.414**\$\pm\$35.593
 TrkWithGPSDat->Opt : 7.8722\$\pm\$4.5505
 TrkWithGPSDat->Outl : 0.08174\$\pm\$0.048233
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.5143\$\pm\$8.6564
New KF decision: 0.26582\$\pm\$0.3606
Total Tracking: nan\$\pm\$nan

Ejecución 4

ORB Extraction: 148.23\$\pm\$11.251
Pose Prediction: **23.133**\$\pm\$30.702
MAPEO->ChkRplInLstFrm: 0.12319\$\pm\$0.058022
MAPEO->TrkWithGPSDat : 46.59\$\pm\$28.236
 TrkWithGPSDat->Header : 44.816\$\pm\$23.879
 TrkWithGPSDat->HeaderSrch : **40.812**\$\pm\$25.759
 TrkWithGPSDat->Opt : 5.7295\$\pm\$2.9257
 TrkWithGPSDat->Outl : 0.057183\$\pm\$0.022449
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 3.8989\$\pm\$5.0192
New KF decision: 0.14151\$\pm\$0.27151
Total Tracking: nan\$\pm\$nan

Ejecución 5

ORB Extraction: 144.53\$\pm\$11.805
Pose Prediction: **12.841**\$\pm\$15.617
MAPEO->ChkRplInLstFrm: 0.079383\$\pm\$0.038036
MAPEO->TrkWithGPSDat : 20.839\$\pm\$15.05
 TrkWithGPSDat->Header : 20.001\$\pm\$12.811
 TrkWithGPSDat->HeaderSrch : **17.501**\$\pm\$13.442
 TrkWithGPSDat->Opt : 3.2435\$\pm\$1.9463
 TrkWithGPSDat->Outl : 0.039076\$\pm\$0.011862
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 2.7239\$\pm\$3.0615
New KF decision: 0.15302\$\pm\$0.28425
Total Tracking: nan\$\pm\$nan

Ejecución 6

ORB Extraction: 144.61\$\pm\$12.027
Pose Prediction: **35.993**\$\pm\$39.979
MAPEO->ChkRplInLstFrm: 0.14915\$\pm\$0.08424
MAPEO->TrkWithGPSDat : 54.217\$\pm\$37.527
 TrkWithGPSDat->Header : 51.537\$\pm\$31.698
 TrkWithGPSDat->HeaderSrch : **47.206**\$\pm\$33.323
 TrkWithGPSDat->Opt : 7.2155\$\pm\$4.3995
 TrkWithGPSDat->Outl : 0.070218\$\pm\$0.038777
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 7.0409\$\pm\$7.5901
New KF decision: 0.22469\$\pm\$0.33515
Total Tracking: nan\$\pm\$nan

Ejecución 7

ORB Extraction: 146.84\$\pm\$11.172
Pose Prediction: **39.897**\$\pm\$39.507
MAPEO->ChkRplInLstFrm: 0.16282\$\pm\$0.076539
MAPEO->TrkWithGPSDat : 59.888\$\pm\$33.653
 TrkWithGPSDat->Header : 55.443\$\pm\$28.172
 TrkWithGPSDat->HeaderSrch : **52.603**\$\pm\$29.958
 TrkWithGPSDat->Opt : 7.4771\$\pm\$3.7376
 TrkWithGPSDat->Outl : 0.074649\$\pm\$0.037041
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.1014\$\pm\$7.6719
New KF decision: 0.25348\$\pm\$0.33991
Total Tracking: nan\$\pm\$nan

Ejecución 8

ORB Extraction: 148.03\$\pm\$11.121
Pose Prediction: **43.003**\$\pm\$44.593
MAPEO->ChkRplInLstFrm: 0.17113\$\pm\$0.086856
MAPEO->TrkWithGPSDat : 64.552\$\pm\$39.722

TrkWithGPSDat->Header : 59.734\$\pm\$33.88

TrkWithGPSDat->HeaderSrch : **56.71**\$\pm\$35.392

TrkWithGPSDat->Opt : 8.0043\$\pm\$4.4255

TrkWithGPSDat->Outl : 0.08127\$\pm\$0.045874

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 8.7093\$\pm\$8.6734

New KF decision: 0.26274\$\pm\$0.35222

Total Tracking: nan\$\pm\$nan

Estudio Flags de compilación

1) Los proyectos se compilan con un script de compilación que a su vez utiliza cmake (<https://blog.adrianistan.eu/tutorial-de-cmake>)

La secuencia normal de compilación con cmake es:

```
mkdir -p build # Crea una carpeta build, -p para no dar error si ya existe
cmake .. # Genera un Makefile con las opciones por defecto
make # Make sigue las directrices del Makefile para invocar compilador, etc
```

donde la órden cmake genera un Makefile con las opciones especificadas en el archivo CMakeLists.txt

p. ej:

```
cmake_minimum_required(VERSION 2.8)
project(DBoW2)
set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wall -O3 ")
```

cmake puede ejecutarse con las opciones por defecto con:

```
cmake ..
```

o especificando variables intenas como:

```
cmake -DCMAKE_CXX_FLAGS='-std=c++11' \
```

CMake procesa estas definiciones de la siguiente forma:

1. **Desde la línea de comando:**

- CMAKE_CXX_FLAGS se establece en '-std=c++11'.

2. **En el archivo CMakeLists.txt:**

- La línea `set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wall -O3")` se ejecuta.
- Aquí, `${CMAKE_CXX_FLAGS}` se expande al valor actual de `CMAKE_CXX_FLAGS` que fue establecido desde la línea de comando, es decir, `-std=c++11`.

Por lo tanto, el valor final de `CMAKE_CXX_FLAGS` será: `-std=c++11 -Wall -O3`

2) La compilación se realiza desde un script que llama a cmake pasando una serie de flags. Estos flags se concatenan a los que se especifican en el CMakeLists.txt correspondiente (hay un CMakeLists por cada librería y otro para el proyecto vbn) code/

```
/ORB_SLAM3
|-- CMakeLists.txt  project(ORB_SLAM3)
                    set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wall -O3")
                    set(CMAKE_CXX_FLAGS_RELEASE "${CMAKE_CXX_FLAGS_RELEASE}")

/lib
|-- libORB_SLAM3.so

/ThirdParty
/DBow2
|-- CMakeLists.txt  project(DBow2)
                    set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wall -O3")
                    SET(g2o_CXX_FLAGS "${g2o_CXX_FLAGS} -Wall -W")
                    SET(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} ${g2o_CXX_FLAGS}")

/lib
|-- libDBow2.so

/g2o
|-- CMakeLists.txt  project(g2o)
                    SET(CMAKE_CXX_FLAGS_RELEASE "${CMAKE_CXX_FLAGS_RELEASE} -O3")

/lib
|-- libg2o.so

/Sophus
|-- CMakeLists.txt  project(Sophus)
                    SET(CMAKE_CXX_FLAGS_RELEASE "-O3")
                    SET(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wall -Werror -Wextra ....)

/lib
|-- libSophus.so

/project
/cmake
|--- CMakeLists.txt  project(vbn)
                    set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -g")

/build
|-- vbn

/scripts
|--- cross_build_orb_slam3.sh
    echo "Configuring and building DBow2 ..."
    cmake -D CMAKE_BUILD_TYPE=Release
    -DCMAKE_CXX_FLAGS='-std=c++11'
    echo "Configuring and building Sophus ..."
    cmake -D CMAKE_BUILD_TYPE=Release
    -DCMAKE_CXX_FLAGS='-std=c++11'
    echo "Configuring and building g2o ..."
    cmake -D CMAKE_BUILD_TYPE=Release
    -DCMAKE_CXX_FLAGS='-std=c++11'
    echo "Configuring and building ORB_SLAM3 ..."
    cmake -D CMAKE_BUILD_TYPE=Release
    -DCMAKE_CXX_FLAGS='-std=c++11'
|--- cross_build_vbn.sh
    echo "Configuring and building VBN ..."
```

3) Exploración de las opciones que ofrece GNU g++
(<https://gcc.gnu.org/onlinedocs/gcc/AArch64-Options.html>)

- Versión del compilador utilizado:

```
aarch64-linux-gnu-g++ --version  
aarch64-linux-gnu-g++ (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0
```

- Arquitectura CPU:

```
cat /proc/cpuinfo  
processor      : 0  
BogoMIPS      : 60.00  
Features       : fp asimd evtstrm aes pmull sha1 sha2 crc32 cpuid  
CPU implementer : 0x41  
CPU architecture: 8  
CPU variant    : 0x0  
CPU part       : 0xd03  
CPU revision   : 4
```

Estudio compilación g2o: los flags deben cambiarse en el CMakeLists.txt

```
SET(CMAKE_CXX_FLAGS_RELEASE "${CMAKE_CXX_FLAGS_RELEASE} -O0")
```

- -O3 -march=armv8.3-a
- -O3 -mtune=cortex-a53
- -O3 -march=armv8.3-a -mtune=cortex-a53
- -O3 -mfpu=neon-vfpv3 error al compilar no se admite en la versión de g++
- -O3, -O2, -O0
- Se intentó deshabilitar la vectorización según indica:
<https://eigen.tuxfamily.org/index.php?title=FAQ#Vectorization>
se añade `add_definitions(-DEIGEN_DONT_VECTORIZE=1)` al CMakeLists.txt
el ejecutable es ligeramente distinto pero siguen apareciendo instrucciones neon
"Also notice that your compiler may still be auto-vectorizing."
- -O3 -march= armv8.3-a+nosimd


```
-rwxr-xr-x 1 root  root  16104424 jun 11 14:38 libg2o_O0.so  
-rwxr-xr-x 1 root  root   672136 jun 11 14:39 libg2o_O2.so  
-rwxr-xr-x 1 root  root   663712 jun 11 22:28 libg2o_O3_armv8.3-a+nosimd.so  
-rwxr-xr-x 1 root  root   693032 jun 11 22:26 libg2o_O3_armv8.3-a.so  
-rwxr-xr-x 1 root  root   697128 jun 11 22:16 libg2o_O3_mtune.so  
-rwxr-xr-x 1 root  root   671856 jun 11 21:41 libg2o_O3_NoVect.so  
-rwxr-xr-x 1 root  root   693032 may 29 13:50 libg2o_O3.so
```
- Se desensamblaron -O0, -O2, -O3, march=armv8.3-a, -mtune=cortex-a53, -march= armv8.3-a+nosimd identificando instrucciones neon (fdiv, fmul, con registros Vx) en todos los ejecutables excepto en **armv8.3-a+nosimd**

Resultados de los test

- frecuencia del procesador

```
cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq  
1199999
```

ligg2o_00.so

ORB Extraction: 145.89\$\pm\$12.575
Pose Prediction: **73.312**\$\pm\$50.721
MAPEO->ChkRplInLstFrm: 0.18626\$\pm\$0.07878
MAPEO->TrkWithGPSDat : **86.997**\$\pm\$42.898
 TrkWithGPSDat->Header : 63.133\$\pm\$30.593
 TrkWithGPSDat->HeaderSrch : **62.842**\$\pm\$30.786
 TrkWithGPSDat->Opt : **24.119**\$\pm\$12.417
 TrkWithGPSDat->Outl : 0.096403\$\pm\$0.045858
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 27.727\$\pm\$18.672
New KF decision: 0.33384\$\pm\$0.33419
Total Tracking: nan\$\pm\$nan

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ORB Extraction: 145.3\$\pm\$12.377
Pose Prediction: **71.772**\$\pm\$48.777
LM Track: 27.106\$\pm\$17.731
New KF decision: 0.32589\$\pm\$0.32723
Total Tracking: nan\$\pm\$nan

ORB Extraction: 147.25\$\pm\$11.132
Pose Prediction: **73.699**\$\pm\$50.962
MAPEO->ChkRplInLstFrm: 0.17815\$\pm\$0.074737
MAPEO->TrkWithGPSDat : 87.418\$\pm\$43.101
 TrkWithGPSDat->Header : 63.508\$\pm\$30.658
 TrkWithGPSDat->HeaderSrch : **63.215**\$\pm\$30.853
 TrkWithGPSDat->Opt : 24.164\$\pm\$12.592
 TrkWithGPSDat->Outl : 0.097971\$\pm\$0.046437
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 27.89\$\pm\$18.815
New KF decision: 0.33202\$\pm\$0.33322
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.65\$\pm\$12.063
Pose Prediction: **73.997**\$\pm\$53.033

MAPEO->ChkRplInLstFrm: 0.18013\$\pm\$0.079267
MAPEO->TrkWithGPSDat : 87.791\$\pm\$45.864
 TrkWithGPSDat->Header : 63.632\$\pm\$32.665
 TrkWithGPSDat->HeaderSrch : **63.341**\$\pm\$32.835
 TrkWithGPSDat->Opt : 24.408\$\pm\$13.319
 TrkWithGPSDat->Outl : 0.098172\$\pm\$0.049452
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 28.009\$\pm\$19.581
New KF decision: 0.33845\$\pm\$0.34094
Total Tracking: nan\$\pm\$nan

ORB Extraction: 145.75\$\pm\$12.479
Pose Prediction: **73.827**\$\pm\$51.294
MAPEO->ChkRplInLstFrm: 0.18265\$\pm\$0.075986
MAPEO->TrkWithGPSDat : 87.605\$\pm\$43.51
 TrkWithGPSDat->Header : 63.569\$\pm\$31.127
 TrkWithGPSDat->HeaderSrch : **63.277**\$\pm\$31.314
 TrkWithGPSDat->Opt : 24.285\$\pm\$12.507
 TrkWithGPSDat->Outl : 0.098391\$\pm\$0.047079
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 27.93\$\pm\$18.842
New KF decision: 0.33591\$\pm\$0.33572
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.36\$\pm\$11.72
Pose Prediction: **73.455**\$\pm\$52.291
MAPEO->ChkRplInLstFrm: 0.17946\$\pm\$0.076243
MAPEO->TrkWithGPSDat : 87.17\$\pm\$45.03
 TrkWithGPSDat->Header : 63.385\$\pm\$32.242
 TrkWithGPSDat->HeaderSrch : **63.091**\$\pm\$32.422
 TrkWithGPSDat->Opt : 24.043\$\pm\$12.915
 TrkWithGPSDat->Outl : 0.098117\$\pm\$0.047846
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 27.569\$\pm\$19.052
New KF decision: 0.33329\$\pm\$0.33237
Total Tracking: nan\$\pm\$nan

libg2o_O2.so

ORB Extraction: 144.53\$\pm\$12.015
Pose Prediction: **38.474**\$\pm\$40.544
MAPEO->ChkRplInLstFrm: 0.15442\$\pm\$0.078378
MAPEO->TrkWithGPSDat : 57.728\$\pm\$36.604
 TrkWithGPSDat->Header : 56.106\$\pm\$29.658

TrkWithGPSDat->HeaderSrch : 50.679\$\pm\$32.611
TrkWithGPSDat->Opt : 7.5557\$\pm\$3.8559
TrkWithGPSDat->Outl : 0.076813\$\pm\$0.038573
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 7.8242\$\pm\$7.803
New KF decision: 0.23506\$\pm\$0.32591
Total Tracking: nan\$\pm\$nan

ORB Extraction: 143.28\$\pm\$11.568
Pose Prediction: **23.698**\$\pm\$26.364
MAPEO->ChkRplInLstFrm: 0.11175\$\pm\$0.059897
MAPEO->TrkWithGPSDat : 35.471\$\pm\$24.833
TrkWithGPSDat->Header : 33.155\$\pm\$20.969
TrkWithGPSDat->HeaderSrch : 30.525\$\pm\$21.852
TrkWithGPSDat->Opt : 4.9766\$\pm\$3.2093
TrkWithGPSDat->Outl : 0.052028\$\pm\$0.026643
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.9205\$\pm\$5.3992
New KF decision: 0.20734\$\pm\$0.33828
Total Tracking: nan\$\pm\$nan

ORB Extraction: 143.48\$\pm\$11.181
Pose Prediction: **25.337**\$\pm\$27.393
MAPEO->ChkRplInLstFrm: 0.1157\$\pm\$0.06109
MAPEO->TrkWithGPSDat : 37.916\$\pm\$25.281
TrkWithGPSDat->Header : 35.98\$\pm\$21.258
TrkWithGPSDat->HeaderSrch : 32.442\$\pm\$22.534
TrkWithGPSDat->Opt : 5.4058\$\pm\$3.1963
TrkWithGPSDat->Outl : 0.054934\$\pm\$0.025693
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 5.1973\$\pm\$5.4275
New KF decision: 0.20883\$\pm\$0.3276
Total Tracking: nan\$\pm\$nan

ORB Extraction: 145.02\$\pm\$12.263
Pose Prediction: **13.28**\$\pm\$15.764
MAPEO->ChkRplInLstFrm: 0.08239\$\pm\$0.03688
MAPEO->TrkWithGPSDat : 22.208\$\pm\$14.657
TrkWithGPSDat->Header : 20.454\$\pm\$12.508
TrkWithGPSDat->HeaderSrch : 18.706\$\pm\$13.018
TrkWithGPSDat->Opt : 3.3171\$\pm\$1.9699
TrkWithGPSDat->Outl : 0.039313\$\pm\$0.012655
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 2.8234\$\pm\$3.1436
New KF decision: 0.15091\$\pm\$0.27803
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.2\$\pm\$11.492

Pose Prediction: 7.1359\$\pm\$14.748

MAPEO->ChkRplInLstFrm: 0.083598\$\pm\$0.048342
MAPEO->TrkWithGPSDat : 24.117\$\pm\$17.967
 TrkWithGPSDat->Header : 26.189\$\pm\$14.718
 TrkWithGPSDat->HeaderSrch : 19.498\$\pm\$16.368
 TrkWithGPSDat->Opt : 4.1785\$\pm\$2.4105
 TrkWithGPSDat->Outl : 0.044492\$\pm\$0.016017
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 1.3456\$\pm\$2.8226
New KF decision: 0.083183\$\pm\$0.22947
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.41\$\pm\$11.434

Pose Prediction: **39.054**\$\pm\$40.69

MAPEO->ChkRplInLstFrm: 0.15724\$\pm\$0.079558
MAPEO->TrkWithGPSDat : 58.682\$\pm\$36.39
 TrkWithGPSDat->Header : 54.557\$\pm\$31.03
 TrkWithGPSDat->HeaderSrch : 51.471\$\pm\$32.514
 TrkWithGPSDat->Opt : 7.3436\$\pm\$4.0263
 TrkWithGPSDat->Outl : 0.074884\$\pm\$0.039509
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 7.8081\$\pm\$7.6762
New KF decision: 0.24138\$\pm\$0.32687
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.52\$\pm\$11.956

Pose Prediction: **40.58**\$\pm\$41.19

MAPEO->ChkRplInLstFrm: 0.15826\$\pm\$0.074898
MAPEO->TrkWithGPSDat : 60.827\$\pm\$35.974
 TrkWithGPSDat->Header : 56.551\$\pm\$30.395
 TrkWithGPSDat->HeaderSrch : 53.502\$\pm\$32.1
 TrkWithGPSDat->Opt : 7.5012\$\pm\$3.9496
 TrkWithGPSDat->Outl : 0.076524\$\pm\$0.039695
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.1591\$\pm\$7.8897
New KF decision: 0.25181\$\pm\$0.33646
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.09\$\pm\$11.377

Pose Prediction: 12.609\$\pm\$14.917

MAPEO->ChkRplInLstFrm: 0.075771\$\pm\$0.038523

MAPEO->TrkWithGPSDat : 19.329\$\pm\$14.463

TrkWithGPSDat->Header : 18.502\$\pm\$12.639

TrkWithGPSDat->HeaderSrch : 15.987\$\pm\$12.99

TrkWithGPSDat->Opt : 3.0016\$\pm\$1.9787

TrkWithGPSDat->Outl : 0.037811\$\pm\$0.013106

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 2.6629\$\pm\$2.9907

New KF decision: 0.15642\$\pm\$0.28308

Total Tracking: nan\$\pm\$nan

libg2o_O3.so

ORB Extraction: 145.53\$\pm\$12.281

Pose Prediction: **42.077**\$\pm\$42.552

MAPEO->ChkRplInLstFrm: 0.18128\$\pm\$0.085378

MAPEO->TrkWithGPSDat : 63.157\$\pm\$37.026

TrkWithGPSDat->Header : 58.535\$\pm\$31.341

TrkWithGPSDat->HeaderSrch : **55.543**\$\pm\$33.049

TrkWithGPSDat->Opt : **7.8008**\$\pm\$4.0311

TrkWithGPSDat->Outl : 0.076764\$\pm\$0.038593

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 8.4274\$\pm\$8.1059

New KF decision: 0.25286\$\pm\$0.33743

Total Tracking: nan\$\pm\$nan

RB Extraction: 148.13\$\pm\$10.872

Pose Prediction: **41.296**\$\pm\$43.791

MAPEO->ChkRplInLstFrm: 0.16543\$\pm\$0.08687

MAPEO->TrkWithGPSDat : 61.731\$\pm\$39.881

TrkWithGPSDat->Header : 58.659\$\pm\$33.426

TrkWithGPSDat->HeaderSrch : **54.203**\$\pm\$35.559

TrkWithGPSDat->Opt : **7.8532**\$\pm\$4.3641

TrkWithGPSDat->Outl : 0.078395\$\pm\$0.043404

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 8.3677\$\pm\$8.4912

New KF decision: 0.25965\$\pm\$0.35692

Total Tracking: nan\$\pm\$nan

ORB Extraction: 147\$\pm\$11.143
Pose Prediction: **23.23**\$\pm\$25.012
MAPEO->ChkRplInLstFrm: 0.10867\$\pm\$0.055755
MAPEO->TrkWithGPSDat : 34.758\$\pm\$22.994
 TrkWithGPSDat->Header : 32.214\$\pm\$19.21
 TrkWithGPSDat->HeaderSrch : **30.008**\$\pm\$20.138
 TrkWithGPSDat->Opt : **4.847**\$\pm\$2.9918
 TrkWithGPSDat->Outl : 0.049352\$\pm\$0.022986
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.7977\$\pm\$5.0485
New KF decision: 0.19703\$\pm\$0.31386
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.12\$\pm\$11.429
Pose Prediction: **22.295**\$\pm\$25.89
MAPEO->ChkRplInLstFrm: 0.11938\$\pm\$0.071564
MAPEO->TrkWithGPSDat : 34.825\$\pm\$24.587
 TrkWithGPSDat->Header : 34.023\$\pm\$20.135
 TrkWithGPSDat->HeaderSrch : **29.844**\$\pm\$21.7
 TrkWithGPSDat->Opt : **5.2217**\$\pm\$3.0663
 TrkWithGPSDat->Outl : 0.052744\$\pm\$0.025055
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.7078\$\pm\$5.2644
New KF decision: 0.19258\$\pm\$0.32509
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.69\$\pm\$10.398
Pose Prediction: **19.193**\$\pm\$23.725
MAPEO->ChkRplInLstFrm: 0.098655\$\pm\$0.059702
MAPEO->TrkWithGPSDat : 29.949\$\pm\$23.496
 TrkWithGPSDat->Header : 30.35\$\pm\$19.531
 TrkWithGPSDat->HeaderSrch : **25.57**1\$\pm\$20.813
 TrkWithGPSDat->Opt : **4.6308**\$\pm\$2.9602
 TrkWithGPSDat->Outl : 0.047436\$\pm\$0.021693
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.8369\$\pm\$4.6474
New KF decision: 0.16607\$\pm\$0.29165
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.19\$\pm\$11.362
Pose Prediction: **19.842**\$\pm\$24.689
MAPEO->ChkRplInLstFrm: 0.10157\$\pm\$0.057159
MAPEO->TrkWithGPSDat : 31.818\$\pm\$24.312

TrkWithGPSDat->Header : 30.975\$\pm\$20.61
TrkWithGPSDat->HeaderSrch : 27.235\$\pm\$21.56
TrkWithGPSDat->Opt : 4.6573\$\pm\$3.0791
TrkWithGPSDat->Outl : 0.048696\$\pm\$0.024236
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.916\$\pm\$4.7708
New KF decision: 0.17082\$\pm\$0.29338
Total Tracking: nan\$\pm\$nan

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ORB Extraction: 147.14\$\pm\$12.175
Pose Prediction: **19.181**\$\pm\$23.324
LM Track: 3.5062\$\pm\$4.1553
New KF decision: 0.14973\$\pm\$0.27704
Total Tracking: nan\$\pm\$nan

ORB Extraction: 147.56\$\pm\$11.109
Pose Prediction: **15.719**\$\pm\$18.557
LM Track: 3.2763\$\pm\$3.6141
New KF decision: 0.17017\$\pm\$0.30216
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.66\$\pm\$11.987
Pose Prediction: **20.173**\$\pm\$24.129
LM Track: 3.692\$\pm\$4.3158
New KF decision: 0.15111\$\pm\$0.2755
Total Tracking: nan\$\pm\$nan

libg2o_O3.so + vbn -O3

ORB Extraction: 145.4\$\pm\$10.661
Pose Prediction: **35.549**\$\pm\$35.347
MAPEO->ChkRpllnLstFrm: 0.14573\$\pm\$0.06448
MAPEO->TrkWithGPSDat : 53.539\$\pm\$30.128
TrkWithGPSDat->Header : 48.756\$\pm\$26.004
TrkWithGPSDat->HeaderSrch : 47.219\$\pm\$26.901
TrkWithGPSDat->Opt : 6.3322\$\pm\$3.3711
TrkWithGPSDat->Outl : 0.064801\$\pm\$0.029899
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 6.9884\$\pm\$6.4857
New KF decision: 0.24457\$\pm\$0.33549
Total Tracking: nan\$\pm\$nan

ORB Extraction: 147.7\$\pm\$11.228
Pose Prediction: **19.24**\$\pm\$23.956

MAPEO->ChkRplInLstFrm: 0.10112\$\pm\$0.05137
MAPEO->TrkWithGPSDat : 33.89\$\pm\$22.559
 TrkWithGPSDat->Header : 34.143\$\pm\$18.151
 TrkWithGPSDat->HeaderSrch : 29.273\$\pm\$20.355
 TrkWithGPSDat->Opt : 4.8269\$\pm\$2.4823
 TrkWithGPSDat->Outl : 0.048557\$\pm\$0.017359
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.5207\$\pm\$4.2458
New KF decision: 0.15089\$\pm\$0.27781
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.83\$\pm\$11.711
Pose Prediction: **28.688**\$\pm\$30.713
MAPEO->ChkRplInLstFrm: 0.12609\$\pm\$0.066695
MAPEO->TrkWithGPSDat : 43.085\$\pm\$28.076
 TrkWithGPSDat->Header : 40.392\$\pm\$23.582
 TrkWithGPSDat->HeaderSrch : 37.494\$\pm\$24.853
 TrkWithGPSDat->Opt : 5.7127\$\pm\$3.4472
 TrkWithGPSDat->Outl : 0.058085\$\pm\$0.029773
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 5.9097\$\pm\$6.1195
New KF decision: 0.23052\$\pm\$0.35064
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.61\$\pm\$11.5
Pose Prediction: **41.493**\$\pm\$42.707
MAPEO->ChkRplInLstFrm: 0.16522\$\pm\$0.083073
MAPEO->TrkWithGPSDat : 62.265\$\pm\$37.766
 TrkWithGPSDat->Header : 57.658\$\pm\$32.165
 TrkWithGPSDat->HeaderSrch : 54.721\$\pm\$33.703
 TrkWithGPSDat->Opt : 7.7096\$\pm\$4.1423
 TrkWithGPSDat->Outl : 0.080003\$\pm\$0.043589
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.4161\$\pm\$8.2895
New KF decision: 0.26084\$\pm\$0.34898
Total Tracking: nan\$\pm\$nan

lbg2o_O3_armv8.3-a+nosimd

ORB Extraction: 143.85\$\pm\$11.877
Pose Prediction: **41.733**\$\pm\$42.187
MAPEO->ChkRplInLstFrm: 0.16705\$\pm\$0.082745
MAPEO->TrkWithGPSDat : 62.65\$\pm\$36.687
 TrkWithGPSDat->Header : 58.039\$\pm\$31.06

TrkWithGPSDat->HeaderSrch : **55.098**\$\pm\$32.714
TrkWithGPSDat->Opt : **7.6999**\$\pm\$4.0953
TrkWithGPSDat->Outl : 0.079015\$\pm\$0.041255
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.3189\$\pm\$8.0552
New KF decision: 0.25475\$\pm\$0.34432
Total Tracking: nan\$\pm\$nan

ORB Extraction: 147.35\$\pm\$11.023
Pose Prediction: **44.214**\$\pm\$44.144
MAPEO->ChkRpllnLstFrm: 0.17416\$\pm\$0.079059
MAPEO->TrkWithGPSDat : 66.546\$\pm\$37.845
TrkWithGPSDat->Header : 59.901\$\pm\$33.128
TrkWithGPSDat->HeaderSrch : **58.587**\$\pm\$33.842
TrkWithGPSDat->Opt : **7.9281**\$\pm\$4.1444
TrkWithGPSDat->Outl : 0.077448\$\pm\$0.041321
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.5111\$\pm\$8.0645
New KF decision: 0.23528\$\pm\$0.31729
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.88\$\pm\$11.694
Pose Prediction: **42.313**\$\pm\$44.005
MAPEO->ChkRpllnLstFrm: 0.1682\$\pm\$0.084356
MAPEO->TrkWithGPSDat : 63.437\$\pm\$39.321
TrkWithGPSDat->Header : 58.986\$\pm\$33.564
TrkWithGPSDat->HeaderSrch : **55.816**\$\pm\$35.139
TrkWithGPSDat->Opt : **7.8034**\$\pm\$4.2736
TrkWithGPSDat->Outl : 0.077257\$\pm\$0.041776
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.3876\$\pm\$8.2875
New KF decision: 0.25658\$\pm\$0.34279
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.4\$\pm\$11.672
Pose Prediction: **38.303**\$\pm\$40.459
MAPEO->ChkRpllnLstFrm: 0.15644\$\pm\$0.080883
MAPEO->TrkWithGPSDat : 57.449\$\pm\$36.651
TrkWithGPSDat->Header : 54.547\$\pm\$30.788
TrkWithGPSDat->HeaderSrch : **50.41**\$\pm\$32.785
TrkWithGPSDat->Opt : **7.288**\$\pm\$3.9725
TrkWithGPSDat->Outl : 0.072791\$\pm\$0.038577
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 7.6346\$\pm\$7.6713
New KF decision: 0.23312\$\pm\$0.32184
Total Tracking: nan\$\pm\$nan

ORB Extraction: 143.21\$\pm\$12.303
Pose Prediction: **39.756**\$\pm\$40.932
MAPEO->ChkRplInLstFrm: 0.15877\$\pm\$0.080554
MAPEO->TrkWithGPSDat : 59.393\$\pm\$36.365
 TrkWithGPSDat->Header : 55.668\$\pm\$30.718
 TrkWithGPSDat->HeaderSrch : 52.155\$\pm\$32.546
 TrkWithGPSDat->Opt : 7.4445\$\pm\$3.9063
 TrkWithGPSDat->Outl : 0.074429\$\pm\$0.038141
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 8.1051\$\pm\$7.8071
New KF decision: 0.24494\$\pm\$0.33094
Total Tracking: nan\$\pm\$nan

ligg2o_Ofast.so

ORB Extraction: 147.79\$\pm\$11.446
Pose Prediction: **20.226**\$\pm\$26.432
MAPEO->ChkRplInLstFrm: 0.12026\$\pm\$0.055248
MAPEO->TrkWithGPSDat : 39.473\$\pm\$24.383
 TrkWithGPSDat->Header : 36.021\$\pm\$20.787
 TrkWithGPSDat->HeaderSrch : **34.457**\$\pm\$21.537
 TrkWithGPSDat->Opt : **5.0963**\$\pm\$2.9348
 TrkWithGPSDat->Outl : 0.053392\$\pm\$0.026549
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.2025\$\pm\$5.2211
New KF decision: 0.16728\$\pm\$0.30672
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.56\$\pm\$12.228
Pose Prediction: **23.862**\$\pm\$31.025
MAPEO->ChkRplInLstFrm: 0.1255\$\pm\$0.059242
MAPEO->TrkWithGPSDat : 47.075\$\pm\$28.257
 TrkWithGPSDat->Header : 45.166\$\pm\$23.789
 TrkWithGPSDat->HeaderSrch : **41.35**\$\pm\$25.683
 TrkWithGPSDat->Opt : **5.7388**\$\pm\$2.9257
 TrkWithGPSDat->Outl : 0.056642\$\pm\$0.022667
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.0201\$\pm\$5.0208
New KF decision: 0.14994\$\pm\$0.30016
Total Tracking: nan\$\pm\$nan

ORB Extraction: 143.78\$\pm\$11.474
Pose Prediction: **24.152**\$\pm\$31.359
MAPEO->ChkRplInLstFrm: 0.12842\$\pm\$0.058235
MAPEO->TrkWithGPSDat : 48.432\$\pm\$28.051
 TrkWithGPSDat->Header : 47.437\$\pm\$22.942
 TrkWithGPSDat->HeaderSrch : **42.266**\$\pm\$25.781
 TrkWithGPSDat->Opt : **6.1112**\$\pm\$2.8419
 TrkWithGPSDat->Outl : 0.059638\$\pm\$0.02361
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.1392\$\pm\$5.2288
New KF decision: 0.14692\$\pm\$0.27901
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.83\$\pm\$12.125
Pose Prediction: **20.435**\$\pm\$26.824
MAPEO->ChkRplInLstFrm: 0.11793\$\pm\$0.054858
MAPEO->TrkWithGPSDat : 39.763\$\pm\$24.987
 TrkWithGPSDat->Header : 36.572\$\pm\$21.385
 TrkWithGPSDat->HeaderSrch : **34.842**\$\pm\$22.191
 TrkWithGPSDat->Opt : **5.0037**\$\pm\$2.9089
 TrkWithGPSDat->Outl : 0.05271\$\pm\$0.026373
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.1193\$\pm\$5.1229
New KF decision: 0.16617\$\pm\$0.30394
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.34\$\pm\$11.245
Pose Prediction: **22.064**\$\pm\$30.562
MAPEO->ChkRplInLstFrm: 0.13081\$\pm\$0.059371
MAPEO->TrkWithGPSDat : 48.518\$\pm\$27.622
 TrkWithGPSDat->Header : 45.902\$\pm\$23.292
 TrkWithGPSDat->HeaderSrch : **42.697**\$\pm\$25.12
 TrkWithGPSDat->Opt : **5.845**\$\pm\$2.7694
 TrkWithGPSDat->Outl : 0.057647\$\pm\$0.022957
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.7411\$\pm\$4.9703
New KF decision: 0.12835\$\pm\$0.2606
Total Tracking: nan\$\pm\$nan

ORB Extraction: 147.33\$\pm\$10.853
Pose Prediction: **21.121**\$\pm\$29.222
MAPEO->ChkRplInLstFrm: 0.13088\$\pm\$0.068941

MAPEO->TrkWithGPSDat : 41.272\$\pm\$28.771
TrkWithGPSDat->Header : 38.36\$\pm\$24.968
TrkWithGPSDat->HeaderSrch : **35.787**\$\pm\$25.7
TrkWithGPSDat->Opt : **5.3777**\$\pm\$3.455
TrkWithGPSDat->Outl : 0.057381\$\pm\$0.03249
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.2757\$\pm\$5.6572
New KF decision: 0.16676\$\pm\$0.30451
Total Tracking: nan\$\pm\$nan

ORB Extraction: 146.97\$\pm\$10.972
Pose Prediction: **21.794**\$\pm\$31.096
MAPEO->ChkRpllnLstFrm: 0.12288\$\pm\$0.058262
MAPEO->TrkWithGPSDat : 47.918\$\pm\$29.397
TrkWithGPSDat->Header : **46.788**\$\pm\$24.721
TrkWithGPSDat->HeaderSrch : 41.787\$\pm\$27.065
TrkWithGPSDat->Opt : **5.9727**\$\pm\$2.9756
TrkWithGPSDat->Outl : 0.05835\$\pm\$0.023111
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.6504\$\pm\$5.0749
New KF decision: 0.12419\$\pm\$0.25441
Total Tracking: nan\$\pm\$nan

ORB Extraction: 147.51\$\pm\$10.822
Pose Prediction: **19.25**\$\pm\$25.95
MAPEO->ChkRpllnLstFrm: 0.11519\$\pm\$0.056053
MAPEO->TrkWithGPSDat : 37.641\$\pm\$24.876
TrkWithGPSDat->Header : 34.462\$\pm\$21.416
TrkWithGPSDat->HeaderSrch : **32.957**\$\pm\$22.032
TrkWithGPSDat->Opt : **4.7552**\$\pm\$2.9349
TrkWithGPSDat->Outl : 0.051193\$\pm\$0.026653
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.8977\$\pm\$4.9886
New KF decision: 0.15879\$\pm\$0.29741
Total Tracking: nan\$\pm\$nan

ligg2o_Ofast_armv8.3-a_cortex-a53.so

ORB Extraction: 142.14\$\pm\$11.368
Pose Prediction: **22.62**\$\pm\$31.337
MAPEO->ChkRpllnLstFrm: 0.12668\$\pm\$0.059722
MAPEO->TrkWithGPSDat : 48.184\$\pm\$29.168
TrkWithGPSDat->Header : 46.513\$\pm\$24.181

TrkWithGPSDat->HeaderSrch : 42.334\$\pm\$26.421
TrkWithGPSDat->Opt : 5.9979\$\pm\$2.9586
TrkWithGPSDat->Outl : 0.060131\$\pm\$0.024677
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.8747\$\pm\$5.188
New KF decision: 0.1327\$\pm\$0.26443
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.27\$\pm\$11.386
Pose Prediction: 22.772\$\pm\$31.437
MAPEO->ChkRpllnLstFrm: 0.12446\$\pm\$0.059435
MAPEO->TrkWithGPSDat : 47.467\$\pm\$29.66
TrkWithGPSDat->Header : 46.086\$\pm\$24.795
TrkWithGPSDat->HeaderSrch : 41.695\$\pm\$26.931
TrkWithGPSDat->Opt : 5.8935\$\pm\$2.9769
TrkWithGPSDat->Outl : 0.058773\$\pm\$0.025002
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.8813\$\pm\$5.1218
New KF decision: 0.13397\$\pm\$0.26461
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.24\$\pm\$11.414
Pose Prediction: 25.079\$\pm\$31.473
MAPEO->ChkRpllnLstFrm: 0.12855\$\pm\$0.055334
MAPEO->TrkWithGPSDat : 49.056\$\pm\$27.429
TrkWithGPSDat->Header : 44.975\$\pm\$23.82
TrkWithGPSDat->HeaderSrch : 43.279\$\pm\$24.716
TrkWithGPSDat->Opt : 5.7218\$\pm\$2.9626
TrkWithGPSDat->Outl : 0.058258\$\pm\$0.023972
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 4.2848\$\pm\$5.1757
New KF decision: 0.15358\$\pm\$0.28404
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.58\$\pm\$12.047
Pose Prediction: 20.692\$\pm\$28.85
MAPEO->ChkRpllnLstFrm: 0.12071\$\pm\$0.065641
MAPEO->TrkWithGPSDat : 40.447\$\pm\$28.647
TrkWithGPSDat->Header : 37.247\$\pm\$24.834
TrkWithGPSDat->HeaderSrch : 35.194\$\pm\$25.475
TrkWithGPSDat->Opt : 5.2109\$\pm\$3.4053
TrkWithGPSDat->Outl : 0.055938\$\pm\$0.031882
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 4.1427 $\pm$ 5.5102
New KF decision: 0.16524 $\pm$ 0.30681
Total Tracking: nan $\pm$ nan

limpiar.sh (limpiar caches)

Optimización g2o (el resto de libs y la app con -O3)	libg2o.so size	PosePrediction exec time
-O0	16104424	73 - 71
-O2	672136	40 - 7
-O3 / -O3 -march=armv8.3-a	693032	42 - 19
-O3 dont vectorize	671856	Similar -O3
-O3 -march=armv8.3-a+nosimd	663712	44 - 38
-O3 -march=armv8.3-a -mtune= cortex-a53	697128	25 - 22
-Ofast	689824	24- 19
-Ofast -march=armv8.3-a -mtune= cortex-a53	693920	25 - 20

Libs: -O3 -march=armv8.3-a -mtune=cortex-a53

ORB_SLAM3 , vbn : -O3

ORB Extraction: 146.32\$\pm\$11.763
Pose Prediction: **18.658**\$\pm\$23.928
MAPEO->ChkRplInLstFrm: 0.095665\$\pm\$0.061508
MAPEO->TrkWithGPSDat : 28.724\$\pm\$24.254
 TrkWithGPSDat->Header : 30.218\$\pm\$20.247
 TrkWithGPSDat->HeaderSrch : 24.461\$\pm\$21.454
 TrkWithGPSDat->Opt : 4.6188\$\pm\$3.0907
 TrkWithGPSDat->Outl : 0.048556\$\pm\$0.024595
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.7328\$\pm\$4.6429
New KF decision: 0.16985\$\pm\$0.29338
Total Tracking: nan\$\pm\$nan

ORB Extraction: 148.17\$\pm\$11.019
Pose Prediction: **10.908**\$\pm\$17.963
MAPEO->ChkRplInLstFrm: 0.090794\$\pm\$0.054283
MAPEO->TrkWithGPSDat : 26.087\$\pm\$19.298
 TrkWithGPSDat->Header : 28.144\$\pm\$15.545
 TrkWithGPSDat->HeaderSrch : 21.339\$\pm\$17.559
 TrkWithGPSDat->Opt : 4.5265\$\pm\$2.6169
 TrkWithGPSDat->Outl : 0.046999\$\pm\$0.017794
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 2.2581\$\pm\$3.685
New KF decision: 0.12286\$\pm\$0.26949
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.36\$\pm\$11.954
Pose Prediction: **40.209**\$\pm\$40.244
MAPEO->ChkRplInLstFrm: 0.16349\$\pm\$0.077323
MAPEO->TrkWithGPSDat : 60.308\$\pm\$34.691
 TrkWithGPSDat->Header : 56.029\$\pm\$29.2
 TrkWithGPSDat->HeaderSrch : 53.018\$\pm\$30.973
 TrkWithGPSDat->Opt : 7.465\$\pm\$3.7933
 TrkWithGPSDat->Outl : 0.075956\$\pm\$0.03887
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 7.9489\$\pm\$7.636
New KF decision: 0.2494\$\pm\$0.33168
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.3\$\pm\$11.969
Pose Prediction: **18.487**\$\pm\$24.154
MAPEO->ChkRpllnLstFrm: 0.097366\$\pm\$0.06269
MAPEO->TrkWithGPSDat : 29.541\$\pm\$24.526
 TrkWithGPSDat->Header : 31.094\$\pm\$20.561
 TrkWithGPSDat->HeaderSrch : 25.178\$\pm\$21.841
 TrkWithGPSDat->Opt : 4.6529\$\pm\$3.0393
 TrkWithGPSDat->Outl : 0.048894\$\pm\$0.023559
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.6332\$\pm\$4.6463
New KF decision: 0.16314\$\pm\$0.2864
Total Tracking: nan\$\pm\$nan

ORB Extraction: 144.74\$\pm\$12.177
Pose Prediction: **13.138**\$\pm\$15.899
MAPEO->ChkRpllnLstFrm: 0.074738\$\pm\$0.040037
MAPEO->TrkWithGPSDat : 20.229\$\pm\$15.597
 TrkWithGPSDat->Header : 19.9\$\pm\$13.579
 TrkWithGPSDat->HeaderSrch : 16.628\$\pm\$14.007
 TrkWithGPSDat->Opt : 3.2901\$\pm\$2.1331
 TrkWithGPSDat->Outl : 0.038942\$\pm\$0.013858
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 2.8438\$\pm\$3.2505
New KF decision: 0.16491\$\pm\$0.30188
Total Tracking: nan\$\pm\$nan

ORB Extraction: 142.3\$\pm\$11.431
Pose Prediction: **17.475**\$\pm\$20.607
MAPEO->ChkRpllnLstFrm: 0.093196\$\pm\$0.046282
MAPEO->TrkWithGPSDat : 27.347\$\pm\$19.78
 TrkWithGPSDat->Header : 25.707\$\pm\$17.222
 TrkWithGPSDat->HeaderSrch : 23.293\$\pm\$17.745
 TrkWithGPSDat->Opt : 3.8747\$\pm\$2.4756
 TrkWithGPSDat->Outl : 0.043192\$\pm\$0.018243
MAPEO->TrkRefKeyFrm : nan\$\pm\$nan
MAPEO->Reloc2 : nan\$\pm\$nan
LM Track: 3.6965\$\pm\$3.9813
New KF decision: 0.19014\$\pm\$0.31689
Total Tracking: nan\$\pm\$nan

Optimización todas las libs, orb y vbn	PosePrediction exec time
-O3 -march=armv8.3-a -mtune= cortex-a53	Input sensor was set to: Monocular Illegal instruction Problema con libORB_SLAM3.so
-O3 -march=armv8.3-a -mtune= cortex-a53 (excepto Sophus)	Input sensor was set to: Monocular Illegal instruction
-O3 -march=armv8.3-a -mtune= cortex-a53 (ORB_SLAM3 solo con -mtune)	Frame 794 0 Illegal instruction Illegal instruction Problema con libORB_SLAM3.so
-O3 -march=armv8.3-a -mtune= cortex-a53 (excepto ORB_SLAM3 -O3)	Frame 794 0 Illegal instruction Illegal instruction Problema con libORB_SLAM3.so
-O3 -march=armv8.3-a -mtune= cortex-a53 (excepto ORB_SLAM3 -O3, vbn -O3)	40 - 10

```
int ORBmatcher::SearchByProjection(Frame &F, const vector<MapPoint*>
&vpMapPoints, const float th, const bool bFarPoints, const float thFarPoints)
```

```
{
    for(size_t iMP=0; iMP<vpMapPoints.size(); iMP++) {
        MapPoint* pMP = vpMapPoints[iMP];
        if(pMP->mbTrackInView)
        {
            if(F.Nleft != -1 && pMP->mbTrackInViewR){
            }
        }
        return nmatches;
    }
}
```

REVISAR: en recorded hay unas 12.000 imágenes pero solo se procesan 1.293 frames, de los cuales los primeros 794 parece que dan error

Cambiando los flags

Versión ORBSLAM -02: Tracking

ORB Extraction: 148.01\$\pm\$10.777

Pose Prediction: 99.727\$\pm\$81.502

MAPEO->ChkRplInLstFrm: 0.25194\$\pm\$0.14208

MAPEO->TrkWithGPSDat : 117.78\$\pm\$75.344

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 17.279\$\pm\$13.128

New KF decision: 0.39422\$\pm\$0.41986

Total Tracking: nan\$\pm\$nan

Versión -03 cross: Tracking

ORB Extraction: 148.5\$\pm\$11.31

Pose Prediction: 80.992\$\pm\$66.314

MAPEO->ChkRplInLstFrm: 0.18724\$\pm\$0.11827

MAPEO->TrkWithGPSDat : 90.842\$\pm\$63.335

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 13.319\$\pm\$10.445

New KF decision: 0.33924\$\pm\$0.38297

Total Tracking: nan\$\pm\$nan

Versión -02 cross (-03 full)

Tracking

ORB Extraction: 144.55\$\pm\$12.012

Pose Prediction: 79.854\$\pm\$74.783

MAPEO->ChkRplInLstFrm: 0.20478\$\pm\$0.12889

MAPEO->TrkWithGPSDat : 96.265\$\pm\$71.641

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 13.814\$\pm\$11.842

New KF decision: 0.34132\$\pm\$0.39059

Total Tracking: nan\$\pm\$nan

Versión -O2 cross (-O2 full)

Tracking

ORB Extraction: 148.06\$\pm\$10.721

Pose Prediction: 103.55\$\pm\$87.126

MAPEO->ChkRplInLstFrm: 0.24532\$\pm\$0.13329

MAPEO->TrkWithGPSDat : 122.76\$\pm\$81.201

MAPEO->TrkRefKeyFrm : nan\$\pm\$nan

MAPEO->Reloc2 : nan\$\pm\$nan

LM Track: 17.479\$\pm\$13.261

New KF decision: 0.37798\$\pm\$0.38645

Total Tracking: nan\$\pm\$nan

Versión -O3 cross (-O3 full)

Localización de la función cuello de botella

//ESTIMATE SLAM POSE//

```
slamEstimated = mpSLAM->TrackMonocular(frame, std::stod(records[6]), vision_fix,  
imu_values, "", &Tcw, true, &gTbc, true);
```

TrackMonocular

[https://github.com/embention/DAA/blob/](https://github.com/embention/DAA/blob/5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/ORB_SLAM3/src/System.cc#L390)

[5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/](https://github.com/embention/DAA/blob/5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/ORB_SLAM3/src/System.cc#L390)

[ORB_SLAM3/src/System.cc#L390](https://github.com/embention/DAA/blob/5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/ORB_SLAM3/src/System.cc#L390)

[https://github.com/embention/DAA/blob/](https://github.com/embention/DAA/blob/5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/ORB_SLAM3/src/System.cc#L390)

[23bc1c103ff27dd95aeda7f7abb697981999900e/items/sw_gnssdenied/items/sw_g2o/](https://github.com/embention/DAA/blob/5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/ORB_SLAM3/src/System.cc#L390)

[g2o/core/optimization_algorithm_levenberg.cpp#L43](https://github.com/embention/DAA/blob/5fbb1d7063d69ee69f53360d826d134829d41a82/items/sw_gnssdenied/code/ORB_SLAM3/src/System.cc#L390)

Tracking.cc

```
|-- TrackLocalMap() {  
    if(gmapgps) { // sin GPS  
        PoseOptimizationGPS()  
    }
```

Optimizer.cc

```
|-- PoseOptimizationGPS()
```

La etapa PosePrediction es en realidad la ejecución de la función **PoseOptimizationGPS()**

- Identificación de los casos de uso comunes, descartando los casos excepcionales, donde se ejecuta PoseOptimizationGPS().
(1) con GPS: Map -> TrackLocalMap
(2) sin GPS: Location -> TrackLocalMap

- Identificación de la parte más pesada dentro de la función
PoseOptimizationGPS(): g2o::OptimizationAlgorithmLevenberg(solver_ptr);)

```
// file: Optimizer.cc
int Optimizer::PoseOptimizationGPS(Frame *pFrame){
    solver = new g2o::OptimizationAlgorithmLevenberg(solver_ptr);
    optimizer.setAlgorithm(solver);

    for(size_t it=0; it<4; ++it) {
        optimizer.optimize(its[it]);
    }
}
```

Para confirmar esta hipótesis, sin necesidad de utilizar Gprof, se propone utilizar contadores de tiempo para registrar el tiempo de ejecución del bucle y comparar con el tiempo de ejecución de PoseOptimizationGPS().

Se modifica el main para instrumentarlo y eliminar el COPRO. Se realiza la compilación cruzada y se carga en la placa Genesys para medir los tiempos

****B.2)**** En ocasiones la optimización -O3, la que estamos utilizando actualmente, aumenta el tamaño del código (p. ej por desenrollamiento de bucles) y provoca más fallos de caché ralentizando la ejecución.

Compilamos con O2 para ver el efecto:

- ****Versión -O2**** (app y libs compiladas con -O2): Tracking

```
ORB Extraction: 148.01$\pm$10.777
**Pose Prediction: 99.727$\pm$81.502**
MAPEO->ChkRpllnLstFrm: 0.25194$\pm$0.14208
**MAPEO->TrkWithGPSDat : 117.78$\pm$75.344**
MAPEO->TrkRefKeyFrm : nan$\pm$nan
MAPEO->Reloc2 : nan$\pm$nan
LM Track: 17.279$\pm$13.128
New KF decision: 0.39422$\pm$0.41986
Total Tracking: nan$\pm$nan
```

- ****Versión -O2**** (app y libs compiladas con -O2, script cross -O2): Tracking

```
ORB Extraction: 148.06$\pm$10.721
**Pose Prediction: 103.55$\pm$87.126**
MAPEO->ChkRpllnLstFrm: 0.24532$\pm$0.13329
**MAPEO->TrkWithGPSDat : 122.76$\pm$81.201**
MAPEO->TrkRefKeyFrm : nan$\pm$nan
MAPEO->Reloc2 : nan$\pm$nan
```

LM Track: 17.479\$\pm\$13.261
New KF decision: 0.37798\$\pm\$0.38645
Total Tracking: nan\$\pm\$nan

Observaciones: -O2 ralentiza la ejecución. Es necesario probar otras optimizaciones más agresivas.

4. Opciones de aceleración

Si se confirma la hipótesis, las opciones para acelerar la ejecución de la función son:

a) identificar las librerías que se llaman desde
g2o::OptimizationAlgorithmLevenberg(solver_ptr);
e intentar compilarlas con algún flag de optimización para ARM NEON
<https://eigen.tuxfamily.org/index.php?title=FAQ#Vectorization>

Las opciones de compilación son las siguientes:

```
ORB_SLAM3/
|-- src/CMakeList.txt
    set(CMAKE_C_FLAGS "${CMAKE_C_FLAGS} -Wall -O3")
    set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wall -O3")

|--Thirdparty/g2o/CMakeList.txt
    set(OpenMP_C_FLAGS
"-I/aarch64-rootfs/usr/lib/aarch64-xilinx-linux/9.2.0/include/omp.h")
    set(OpenMP_CXX_FLAGS
"-I/aarch64-rootfs/usr/lib/aarch64-xilinx-linux/9.2.0/include/omp.h")

# Eigen library parallelise itself, though, presumably due to performance issues
# OPENMP is experimental. We experienced some slowdown with it
FIND_PACKAGE(OpenMP)
SET(G2O_USE_OPENMP OFF CACHE BOOL "Build g2o with OpenMP support (EXPERIMENTAL)")
IF(OPENMP_FOUND AND G2O_USE_OPENMP)
    SET(G2O_OPENMP 1)
    SET(g2o_C_FLAGS "${g2o_C_FLAGS} ${OpenMP_C_FLAGS}")
    SET(g2o_CXX_FLAGS "${g2o_CXX_FLAGS} -DEIGEN_DONT_PARALLELIZE ${OpenMP_CXX_FLAGS}")
    MESSAGE(STATUS "Compiling with OpenMP support")
```

```
ENDIF(OPENMP_FOUND AND G2O_USE_OPENMP)
```

```
# Compiler specific options for gcc
```

```
SET(CMAKE_CXX_FLAGS_RELEASE "${CMAKE_CXX_FLAGS_RELEASE} -O3")
```

```
SET(CMAKE_C_FLAGS_RELEASE "${CMAKE_C_FLAGS_RELEASE} -O3")
```

b) ver si tienen opción de acelerar con OpenMP

c) reducir el número de iteraciones del bucle for. Será necesario verificar que funcionalmente la ejecución sigue siendo correcta

d) buscar aceleradores FPGA para esta función

Medidas (short_recorded):

Tracking

ORB Extraction: 160.81 $\pm$ 10.879

Pose Prediction: 81.012 $\pm$ 71.679

LM Track: 12.116 $\pm$ 9.4431

New KF decision: 0.27273 $\pm$ 0.38155

Total Tracking: nan $\pm$ nan

Medidas (recorded):

Tracking

ORB Extraction: 148.58393 $\pm$ 10.70624

Pose Prediction: 76.64363 $\pm$ 67.24267

LM Track: 12.47556 $\pm$ 10.41649

New KF decision: 0.32980 $\pm$ 0.37868

Total Tracking: nan $\pm$ nan

Pasos:

1) confirmar la hipótesis del cuello de botella en

Funciones de Tracking.cc desde la que se llama (frecuencia)

(1) TrackReferenceKeyFrame

(1) TrackWithGPSData

(1) TrackWithMotionModel

(1) TrackLocalMap

(3) Relocalization2

(3) Relocalization

****TODO:**** Tiempo acumulado de PoseOptimizationGPS() en cada una de las funciones

2) cambiar flags de compilación

3) buscar alternativas :

- implementación Hw

- pipeline captura/procesamiento